BLA-05 · DEVELOPER TOOLS AND ML SERVICES

Building It Repeatably, and AI Services You Can Just Use

Infrastructure as code · CloudFormation and the CDK · CLI, SDK and CodeArtifact · Ready-made AI services · Service or foundation model?

PART 03

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Why infrastructure as code?

Clicking in the console	Infrastructure as code
Hard to repeat exactly	Same result every time
No record of what changed	Every change is in version control
Dev and prod slowly drift apart	Dev, test and prod built from one file
Hard to delete everything cleanly	Delete the stack — everything goes

💡 **If you can't rebuild it from a file, you don't really know what you have.**

CloudFormation and the CDK

The CDK is a friendlier way to write CloudFormation.

	AWS CloudFormation	AWS CDK
You write	A template in YAML or JSON	Code in Python, TypeScript and others
Good for	Clear, declarative setups	Logic, loops and reusable building blocks
Under the hood	Creates a stack	Turns your code into CloudFormation

🔁 **Change sets** show what will change before you apply it. ↩️ **Rollback** undoes a stack that fails to create.

CLI, SDK, access keys and CodeArtifact

Tool	Use it for
AWS CLI	Commands in a terminal — quick checks and scripts
AWS SDK (e.g. boto3)	Calling AWS from your own code
Access keys	How the CLI and SDK prove who you are
CodeArtifact	A private, approved store for code packages

⚠️ **Never put access keys in code or on GitHub.** Prefer IAM Identity Center sign-in or roles.

🏢 **CodeArtifact** lets a team use only approved versions of Python and npm packages.

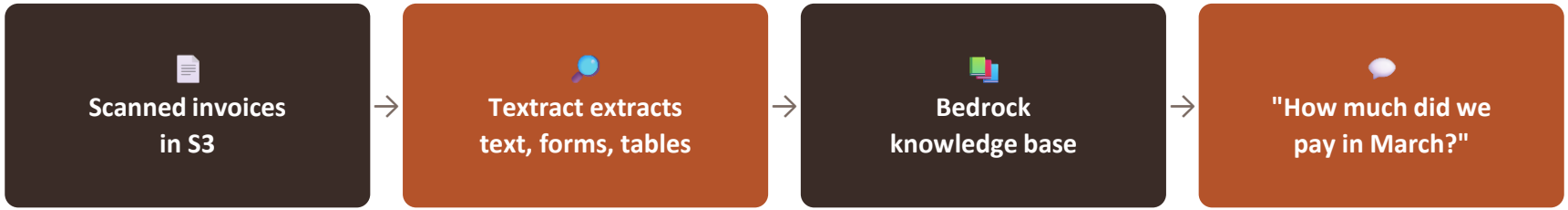
Ready-made AI services

Each does one job well — you don't train anything.

Service	What it does	Example
Rekognition	Understands images and video	Flag unsafe photos in uploads
Textract	Pulls text, forms and tables from documents	Read fields from scanned invoices
Lex	Builds chatbots and voice bots	"Where's my order?" bot
Augmented AI (A2I)	Sends low-confidence results to a person	Human check on unclear invoices
Kendra	Enterprise search	Search across company documents

⚠️ **AWS is retiring A2I and Kendra —** check what replaces them before building on them.

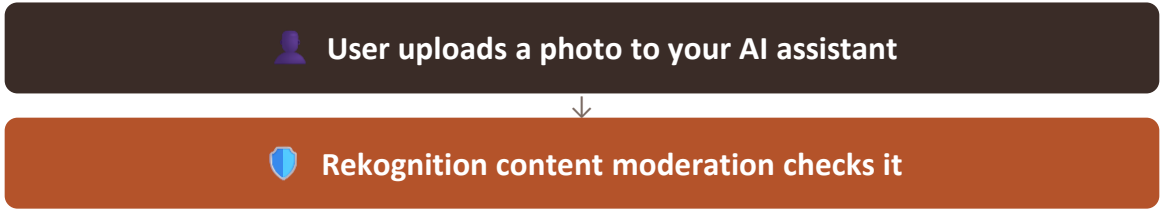
Textract in a GenAI pipeline



Request	Use
One page, small file	Synchronous call — answer straight away
Multi-page PDF	Asynchronous job — start it, then collect the results

Rekognition as a safety check

A separate check that runs before the image reaches the model.



Result	What happens
✅ **Safe**	Send it to the model
❌ **Unsafe**	Reject it
❓ **Unsure**	A person reviews it

💡 **Every result has a confidence score.** You choose the threshold.

Ready-made service or foundation model?

Choose a ready-made service when…	Choose a foundation model when…
The task is fixed and well defined	The task is open-ended
You need the same output format every time	You need flexible, written answers
Volume is high and cost must be predictable	Volume is lower, or quality matters most
"Read the total from this invoice"	"Explain this contract clause to a customer"

💡 **The best apps often use both:** Textract reads the invoice, a model explains it.

Key takeaways

1. If you can't rebuild it from a file, you don't know what you have
2. The CDK is a friendlier way to write CloudFormation
3. Keep access keys out of code — prefer roles and Identity Center
4. Ready-made AI services do one job well and predictably
5. Often the best answer is a service and a model together

Not every AI problem needs a foundation model